

Asymptotically flat metric reconstruction: a mode-based scheme for gravitational self-force calculations

Vahid Toomani,^{*} Stefan Hollands,[†] and Peter Zimmerman[‡]

Abstract

Metric perturbations obtained from a Hertz potential in an ingoing radiation gauge generically grow linearly in r at future null infinity, and are therefore not asymptotically flat in the Bondi sense. Hollands and Toomani showed that this growth is pure gauge and can be removed, within the Green–Hollands–Zimmerman (GHZ) framework, by a residual gauge vector together with a homogeneous, Held-covariantly constant completion of the Hertz potential. In this work we turn that formal construction into a practical, mode-based algorithm. We show that the entire completion is generated by a single Held scalar f° , obtained algebraically from the leading asymptotic Weyl amplitude $\psi_0^{5\circ}$ through a Teukolsky–Starobinsky inversion, and we give explicit (ℓ, m, ω) formulas for the four Held integration constants $\Phi^{0\circ}, \dots, \Phi^{3\circ}$ and for the residual gauge vector ζ^a . In Schwarzschild the construction is diagonal in spin-weighted spherical harmonics and every completion field is available in closed form; in Kerr we give the corresponding (ℓ, m, n, k) scheme, in which the only additional ingredients are the spheroidal-to-spherical mixing coefficients and a tridiagonal $\cos\theta$ coupling matrix. We implement the scheme numerically and demonstrate the cancellation of the growing radiation-gauge terms and the recovery of Bondi falloff. [PZ: Add one closing sentence quoting the actual numerical result once the runs are in, e.g. “the $O(r)$ coefficients cancel to 10^{-N} relative precision for $\ell \leq \ell_{\max}$ ”.]

^{*} vahid.toomani.2@uni-leipzig.de

[†] stefan.hollands@uni-leipzig.de

[‡] zimmerator@protonmail.com

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I. INTRODUCTION

A. Motivation

Extreme-mass-ratio inspirals are among the principal targets of the Laser Interferometer Space Antenna and provide strong motivation for developing accurate perturbative descriptions of compact objects moving in Kerr spacetime [1, 2]. The Teukolsky formalism [3] is particularly well suited to this problem: the extreme Weyl scalars are gauge invariant at linear order and satisfy separable equations, allowing highly accurate calculations in the frequency and time domains. Preserving phase accuracy over an inspiral, however, requires the self-force to post-adiabatic order, and therefore second-order effects, which are sourced

by the metric perturbation itself rather than by the Weyl scalars [4]. A practical scheme for metric reconstruction must therefore provide not only a solution of the linearized Einstein equation, but one with sufficiently good local and asymptotic gauge properties to construct a well defined second order source.

B. GHZ reconstruction and the asymptotic problem of standard radiation gauges

The Green–Hollands–Zimmerman (GHZ) reconstruction formalism [5] extends the Chrzanowski–Cohen–Kegeles–Ori construction [6–9] to the sourced Einstein equation. In the IRG-GHZ puncture scheme, the metric perturbation is decomposed as

$$h_{ab}^{\text{ret}} = x_{ab} + 2 \operatorname{Re} \left(\mathcal{S}_{ab}^{\dagger} \Phi \right) + \dot{g}_{ab} + \mathcal{L}_{\xi} g_{ab}, \quad (\text{I.1})$$

Here Φ is a Hertz potential, satisfying both the $s = -2$ Teukolsky equation and an inversion relation that ties it to the $s = +2$ Weyl scalar ψ_0 ; x_{ab} is a corrector tensor obeying a triangular system of transport equations along the outgoing principal null congruence; and the remaining pieces $\dot{g}_{ab}$ and $\mathcal{L}_{\xi} g_{ab}$ carry the mass, angular-momentum and gauge perturbations [10, 11]. To remove the singular behaviour one further decomposes the retarded field into a puncture and a residual field [12],

$$h_{ab}^{\text{ret}} = h_{ab}^P + h_{ab}^R, \quad (\text{I.2})$$

where the puncture reproduces the local singular structure of the retarded field near the worldline. Multiplying the puncture by a window function renders its effective source of compact radial support.

The asymptotic behavior of the residual perturbation is therefore determined solely by the Hertz potential

$$h_{ab}^R = 2 \operatorname{Re} \left(\mathcal{S}_{ab}^{\dagger} \Phi^R \right), \quad r \gg r_{\text{p}}. \quad (\text{I.3})$$

This leads to an important practical problem. In frequency-domain implementations the Hertz potential is commonly obtained from the computed Weyl scalar ψ_0 using the inversion procedure of Ori [9]. For an outgoing solution at future null infinity,

$$\psi_0 = \frac{\psi_0^{5\circ}(u, \theta, \varphi)}{r^5} + O(r^{-6}), \quad (\text{I.4})$$

whereas the corresponding retarded Hertz potential obtained from the radial inversion

$$\mathbb{P}^4 \bar{\Phi} = 2\psi_0 \quad (\text{I.5})$$

behaves generically as

$$\Phi = O(r^3). \quad (\text{I.6})$$

This growth is familiar from radiation-gauge reconstruction and does not signal a divergence of the physical curvature. It does, however, have an immediate consequence for the reconstructed metric. Applying the second-order reconstruction operator $\mathcal{S}^\dagger$ to such a Hertz potential gives, in the Kinnersley Newman–Penrose frame, the nonvanishing radiation-gauge components schematically as

$$h_{nn}, \quad h_{nm}, \quad h_{mm} = O(r), \quad (\text{I.7})$$

with the corresponding complex-conjugate components. Thus the metric representative obtained directly from the Ori-inverted Hertz potential is not asymptotically flat in the Bondi sense, and the quadratic source $S_{ab}^{(2)}[h, h]$ built from it fails to be integrable against the volume element at $\mathcal{I}^+$, so that neither the second-order Teukolsky source nor the asymptotic flux formulas can be evaluated on it.

C. Asymptotically flat reconstruction

What is needed instead is a representative of the same physical perturbation whose Newman–Penrose components decay at the Bondi rate. Integrability of the second-order source requires at least

$$h_{nn}, \quad h_{nm}, \quad h_{mm} = O(r^{-1}), \quad (\text{I.8})$$

together with

$$h_{ll}, \quad h_{lm}, \quad h_{ln}, \quad h_{m\bar{m}} = O(r^{-2}). \quad (\text{I.9})$$

The discrepancy between the required falloff and what the CCK–Ori procedure delivers is therefore two powers of r in the radiative Newman–Penrose components. **[PZ: Worth stating precisely *where* (I.8)–(I.9) come from — Bondi gauge, or integrability of $S^{(2)}$, or both? Hollands–Toomani Sec. 1.3 and Sec. 3.3.2 have the statement you want.]**

The $O(r)$ behavior is a gauge effect rather than a failure of the Teukolsky solution. Indeed, an asymptotically flat metric may itself be represented using a Hertz potential with an $O(r^3)$ tail, provided that the reconstructed piece is supplemented by the appropriate residual gauge transformation. Hollands and Toomani [13] gave a systematic construction of precisely this transformation. They showed that the unwanted asymptotic terms can be removed by

combining a residual gauge vector ζ^a with a homogeneous modification of the Hertz potential of the form

$$\Phi \longrightarrow \Phi + \Phi^{0\circ} + \frac{\Phi^{1\circ}}{\rho} + \frac{\Phi^{2\circ}}{\rho^2} + \frac{\Phi^{3\circ}}{\rho^3}. \quad (\text{I.10})$$

The apparently growing terms in this homogeneous addition are essential: their contribution under $\mathcal{S}^\dagger$ combines with $\mathcal{L}_\zeta g_{ab}$ to cancel the growing terms in the original radiation-gauge metric, leaving a perturbation with Bondi-type falloff at future null infinity [13].

In this work we turn this formal asymptotic construction into a practical numerical procedure suitable for GHZ puncture calculations. Starting from the asymptotic coefficient $\psi_0^{5\circ}$ of the residual Weyl scalar, we determine the Held scalar f° that generates both the homogeneous Hertz completion and the residual gauge vector. In Schwarzschild spacetime the construction becomes diagonal in spin-weighted spherical harmonics, allowing all completion fields to be obtained explicitly mode by mode. We implement these relations numerically and demonstrate the cancellation of the growing radiation-gauge terms and the recovery of the expected asymptotic falloff. We then extend the construction to Kerr, where the input Teukolsky solution is naturally expressed in spin-weighted spheroidal harmonics, and present numerical results using an m -mode implementation.

The resulting procedure complements the local regularization provided by the Teukolsky puncture scheme. The puncture controls the singular structure in a neighborhood of the compact object, while the asymptotic completion controls the behavior of the reconstructed field at null infinity. Together, these ingredients provide a metric perturbation with substantially better global properties for self-force calculations and, in particular, for the construction of higher-order perturbations in Kerr [spacetime](#).

D. Conventions, notation, and outline

We work in geometrized units $G = c = 1$ and use the metric signature $(+, -, -, -)$. Newman–Penrose and GHP conventions follow GHZ [5], and Held’s modified GHP calculus [14, 15] follows Ref. [13], with $\cdot^\circ$ denoting Held (“ r -independent”) scalars. The Kinnersley tetrad is used throughout, so that

$$\rho = -\frac{1}{r - ia \cos \theta}, \quad \bar{\rho} = -\frac{1}{r + ia \cos \theta}, \quad (\text{I.11})$$

and the nonvanishing background Held scalars are

$$\Psi^\circ = M, \quad \rho^\circ = -\frac{1}{2}, \quad \Omega^\circ = -2ia \cos \theta, \quad (\text{I.12})$$

together with the spin-weight-one Held scalar τ° and its conjugate $\bar{\tau}^\circ$. Note that Ω° is a spin-weight-zero multiplication operator proportional to $\cos \theta$, so it is represented in the ${}_s Y_{\ell,m}$ basis by the same tridiagonal matrix $\mathbf{C}$ that represents ρ^{-1} in (IV.8); explicitly $\Omega^\circ = -2ia \mathbf{C}$.

[PZ: τ° : please insert the explicit form. My earlier guess was wrong and I have removed it rather than substitute another guess; it is a spin-weight-one Held scalar, so whatever it is, its matrix elements have the three-term structure of (IV.5a).]

Four conventions recur and are fixed once here.

- (i) *Signature of the imported Weyl data.* All formulas in this paper are written in the mostly-minus signature $(+, -, -, -)$, as are the GHZ and Held expressions they are built from. Teukolsky amplitudes produced with the Black Hole Perturbation Toolkit [16], however, are in the mostly-plus convention $(-, +, +, +)$, and the two are related by

$$\psi_0^{(+---)} = -\psi_0^{(-+++)}, \quad \Phi^{(+---)} = -\Phi^{(-+++)}. \quad (\text{I.13})$$

The imported amplitudes are therefore negated once, on import; this is the origin of the minus sign in Step 1 of Sec. III F.

- (ii) *Conjugation of a spin-weighted field.* Spin-weighted spherical harmonics obey

$$\overline{{}_s Y_{\ell,m}(\theta, \varphi)} = (-1)^{s+m} {}_{-s} Y_{\ell,-m}(\theta, \varphi), \quad (\text{I.14})$$

so that if X has spin weight s and coefficients $X_{\ell m}$ in the ${}_s Y_{\ell,m}$ basis, then $\bar{X}$ has spin weight $-s$ and its coefficients *in the ${}_{-s} Y_{\ell,m}$ basis* are

$$(\bar{X})_{\ell m} = (-1)^{s+m} \overline{X_{\ell,-m}}, \quad (\text{I.15})$$

where the overline on the right is ordinary complex conjugation of the mode amplitude, accompanied by $\omega \rightarrow -\omega$ (equivalently, by $(m, n, k) \rightarrow (-m, -n, -k)$, since ω_{mnk} is odd under that reflection). The factor $(-1)^s$ is easy to lose because it is unity for every even-spin object in this construction — $\psi_0^{5^\circ}$, $\psi_4^{1^\circ}$, f° and all four Φ^{n° have $s = \pm 2$ — but it equals -1 for the odd-spin ones: ζ_m° , $\bar{\zeta}_m^\circ$, the intermediate chains $\tilde{\delta} f^\circ$ and $\tilde{\delta}' \tilde{\delta}^2 \bar{f}^\circ$, and the metric components h_{nm} , $h_{n\bar{m}}$. See Table I.

- (iii) *Real part.* A real part is meaningful in mode space only for $s = 0$, since X and $\bar{X}$ otherwise live in different harmonic bases. For a real spin-zero scalar, (I.15) gives

$$(\operatorname{Re} X)_{\ell m} = \frac{1}{2} \left(X_{\ell m} + (-1)^m \overline{X_{\ell, -m}} \right), \quad (\text{I.16})$$

which is the projection used for ζ_l° and ζ_n° in (III.13). For a component pair of nonzero spin weight, such as h_{nm} and $h_{n\bar{m}}$, reality of h_{ab} is instead the statement $h_{nm} = \overline{h_{n\bar{m}}}$, which in modes is (I.15) with $s = -1$ and therefore carries an explicit minus sign.

- (iv) *Reconstruction normalization.* We write the reconstructed piece as $h_{ab} = 2 \operatorname{Re}(\mathcal{S}^\dagger \Phi)_{ab}$, so that the operator $\mathcal{S}^\dagger$ quoted in Appendix C is *not* already real-projected. [PZ: This is the one place where the notebook and Vahid's notes differ by a uniform factor of 2 in all four Φ^{n° . Pick one and state it here; the equations below use the notes' normalisation.]

The paper is organized as follows. Section II collects the GHZ decomposition and the Held formalism in the form we need. Section III contains the main analytic result: the adjusted reconstruction, presented as an algorithm generated by a single Held seed f° . Section IV reduces every step of that algorithm to linear algebra on spin-weighted spherical-harmonic coefficients. Sections V and VI present the Schwarzschild and Kerr implementations and numerical results, and Section VII concludes. Appendices collect the Held operator matrix elements, the Teukolsky–Starobinsky constants, and explicit mode expressions for $\mathcal{S}^\dagger$ and $\mathcal{L}_\zeta g$.

II. PRELIMINARIES

A. Kerr background, NP frame, and Held scalars

Held's modification of the GHP formalism [14, 15] is what makes the completion algebraic rather than differential, and we recall only as much of it as we need. In a type-D background one may trade the GHP operators for tilded counterparts $\tilde{\mathbf{P}}', \tilde{\delta}, \tilde{\delta}'$ that commute with radial integration along the outgoing principal null congruence. The general solution of a $\mathbf{P}$ -equation is then a sum of powers of ρ with coefficients that are annihilated by $\mathbf{P}$ — the *Held scalars*, written $\cdot^\circ$ — so that the integration constants of the radial problem become fields on the sphere. The four quantities Φ^{n° of (I.10) are exactly such integration constants, which is why the completion is fixed by angular data alone.

On a fixed Fourier mode $e^{-i\omega u}$ at $\mathcal{I}^+$ the Held operator $\tilde{\mathbf{P}}'$ acts algebraically,

$$\tilde{\mathbf{P}}' \longrightarrow -i\omega, \quad (\text{II.1})$$

which is the single fact that converts the transport equations of Sec. III into algebraic relations. [PZ: State the precise sense of (II.1): $\tilde{\mathbf{P}}'$ is ∂_u acting on the Held (i.e. r -independent) part at $\mathcal{I}^+$.]

The spin weights of every object appearing in the construction are collected in Table I. They fix both the angular matrix elements of Sec. IV C and, through (I.15), the sign attached to each complex conjugation; the odd-spin entries are the ones for which the factor $(-1)^s$ is not unity.

	f°	$\bar{f}^\circ$	Φ^{n°	$\tilde{\delta}\bar{f}^\circ$	$\tilde{\delta}^2\bar{f}^\circ$	$\zeta_l^\circ, \zeta_n^\circ$	ζ_m°	$\bar{\zeta}_m^\circ$
spin weight s	+2	-2	-2	-1	0	0	+1	-1
$(-1)^s$	+	+	+	-	+	+	-	-

	h_{ll}	h_{ln}	h_{nn}	$h_{m\bar{m}}$	h_{nm}	$h_{n\bar{m}}$	h_{mm}
spin weight s	0	0	0	0	+1	-1	+2
$(-1)^s$	+	+	+	+	-	-	+

TABLE I. Spin weights of the fields entering the adjusted reconstruction. The second row of each block is the sign $(-1)^s$ appearing in the conjugation rule (I.15); it is -1 precisely for the odd-spin-weight quantities.

[PZ: The field spin weights are the values hard-coded as `Spinof[...]` in the notebooks. The metric-component spin weights follow from the GHP type of m^a and n^a ; please double-check the sign convention for h_{nm} versus $h_{n\bar{m}}$ against your own, since the two appear swapped between `ReSdagModePhiOps` and `LieZetagComps` (see Appendices C and D).]

B. The GHZ decomposition and the puncture scheme

Following Ori, take a slice Σ of constant Boyer–Lindquist time $t = t_0$ and split the spacetime into the regions to the left and right of the sphere $\mathcal{S}_p$ at $r = r_p$, denoted Σ_- and Σ_+ respectively. In the world-tube region $\Gamma = \{r_{\min} \leq r \leq r_{\max}\}$ define $\Gamma_\pm = \Gamma \cap \Sigma_\pm$, and let the vacuum regions $\Sigma'_\pm$ be the complements of $\Gamma_\pm$ in $\Sigma_\pm$. Define $\Phi^\pm$ to be the regular

(vacuum) solutions supported on $\Sigma'_\pm$, so that

$$h_{ab}^\pm = 2 \operatorname{Re} \mathcal{S}_{ab}^\dagger \Phi^\pm, \quad x \in \Sigma'_\pm. \quad (\text{II.2})$$

The asymptotic decay of h_{ab}^+ , and therefore of the residual field, is completely determined by Φ^+ , which is in turn determined by the decay of ψ_0 .

C. Asymptotics at $\mathcal{I}^+$ and the required falloff

The target of the construction is the pair of conditions (I.8)–(I.9). Two observations make them easier to check. First, in the ingoing radiation gauge the non-radiative components h_{ll} , h_{ln} , h_{lm} , $h_{l\bar{m}}$ and $h_{m\bar{m}}$ vanish *identically*, both for the reconstructed piece (Appendix C) and for the gauge term (Appendix D); condition (I.9) is therefore satisfied trivially, and the whole problem lives in the five radiative components. Second, because the Hertz potential enters through $\mathcal{S}^\dagger$, which is second order in radial derivatives, each radiative component is a finite Laurent polynomial in r near $\mathcal{I}^+$; verifying (I.8) reduces to showing that two coefficients — those of r and of r^0 — vanish in each of them. That is the check carried out in Sec. V C.

[PZ: Decide whether to call the end result “Bondi gauge” or “asymptotically flat gauge”. Strictly you fix the falloff, not the gauge; the residual freedom is the usual BMS supertranslation plus the $\ell = 0, 1$ modes. Hollands–Toomani Sec. 3.3.2 relates Φ^{n° to the news and memory tensors, which would make a nice remark here.]

III. ADJUSTED ASYMPTOTICALLY FLAT GHZ RECONSTRUCTION

A. Asymptotic Weyl amplitudes

We write ${}_s Z_{\ell mnk}^\infty$ and ${}_s Z_{\ell mnk}^H$ for the asymptotic amplitudes of the spin- s Teukolsky solution at future null infinity and at the past horizon respectively; these are precisely the $\mathcal{I}$ and H amplitudes returned by any frequency-domain Teukolsky solver, so that no new radial integrations are needed anywhere in what follows. Only the $\mathcal{I}^+$ amplitudes enter the asymptotic completion. The asymptotic coefficients $\psi_{0,\ell mnk}^{5^\circ} := {}_{+2} Z_{\ell mnk}^\infty$ in

$$\psi_0^{5^\circ}(u, \theta, \varphi) := \sum_{\ell mnk} \psi_{0,\ell mnk}^{5^\circ} {}_{+2} S_{\ell,m}(\theta, \varphi; a\omega_{mnk}) e^{-i\omega_{mnk}u} \quad (\text{III.1})$$

and $\psi_{4,\ell mnk}^{1\circ} := {}_{-2}Z_{\ell mnk}^\infty$ in

$$\psi_4^{1\circ}(u, \theta, \varphi) := \sum_{\ell mnk} \psi_{4,\ell mnk}^{1\circ} {}_{-2}S_{\ell,m}(\theta, \varphi; a\omega_{mnk}) e^{-i\omega_{mnk}u} \quad (\text{III.2})$$

are not independent. The Teukolsky–Starobinsky identities

$$\Delta^2 \mathcal{D}(\Delta^2 \psi_0) = \frac{1}{4} \mathcal{L}^4(\zeta^4 \psi_4) + 3M \partial_t(\zeta^4 \bar{\psi}_4) \quad (\text{III.3})$$

together with the angular inversion relation

$$\mathcal{L}^4 {}_{-2}S_{\ell,m} = D_{\ell m} {}_{+2}S_{\ell,m} \quad (\text{III.4})$$

imply

$${}_{+2}Z_{\ell mnk}^\infty = \frac{1}{4\omega^4} (D_{\ell m} - 12iM\omega p_{\ell mnk}) {}_{-2}Z_{\ell mnk}^\infty, \quad (\text{III.5})$$

where $p_{\ell mnk} = \pm 1$ is a parity factor related to the phase conventions of the spin-weighted spheroidal harmonics. Equivalently, in the form used by the implementation,

$$\psi_{4,\ell m}^{1\circ}(\omega) = \frac{\omega^4}{\mathcal{D}_{\ell m\omega}} \left[{}_{-2}B_{\ell,-m}(-a\omega)^* \psi_{0,\ell m}^{5\circ}(\omega) + 3i\omega(-1)^m \bar{\psi}_{0,\ell,-m}^{5\circ}(-\omega) \right], \quad (\text{III.6})$$

with

$$\mathcal{D}_{\ell m\omega} := {}_{-2}B_{\ell m}(a\omega) \overline{{}_{-2}B_{\ell,-m}(-a\omega)} + 9M^2\omega^2. \quad (\text{III.7})$$

Here $D_{\ell m}$ is the real angular Teukolsky–Starobinsky constant of Appendix B, and ${}_{-2}B_{\ell m}(a\omega)$ is the eigenvalue combination in which the completion formulas are most compactly written. [PZ: These two are the same object up to a factor: in Schwarzschild ${}_2B_\ell = \Lambda_\ell/4$ and $D_\ell = \Lambda_\ell$, so ${}_2B_\ell = D_\ell/4$ and hence $\mathfrak{C}\bar{\mathfrak{C}} = D^2 + 144M^2\omega^2 = 16\mathcal{D}_{\ell m\omega}$. Please confirm that the same identification holds in Kerr; if it does, one of the two symbols can be eliminated from the paper entirely, as we did for $C^\pm$ versus Z .]

For a generic bound geodesic, $\omega_{mnk} = m\Omega_\varphi + n\Omega_r + k\Omega_\theta$ and the polar reflection symmetry of the spheroidal functions gives

$${}_sZ_{\ell,-m,-n,-k}^\infty = (-1)^{\ell+k} ({}_sZ_{\ell mnk}^\infty)^*, \quad (\text{III.8})$$

which halves the number of modes that must be computed. [PZ: The notebooks apply (III.8) to $s = +2$ only; state whether it is meant to hold for general s (it should, being a property of the spheroidal functions under $\theta \rightarrow \pi - \theta$) and cite the source.]

B. The Held seed f°

The entire completion is generated by a single Held scalar f° of spin weight $+2$, defined implicitly by

$$2\tilde{\mathbf{P}}'^3\bar{\psi}_0^{5\circ} = \tilde{\delta}'^4\tilde{\delta}^4\bar{f}^\circ - 9\Psi^\circ\bar{\Psi}^\circ\tilde{\mathbf{P}}'^2\bar{f}^\circ. \quad (\text{III.9})$$

On a Fourier mode, using (II.1), Eq. (III.9) becomes the purely angular relation

$$\tilde{\delta}'^4\tilde{\delta}^4\bar{f}^\circ + 9M^2\omega^2\bar{f}^\circ = 2i\omega^3\bar{\psi}_0^{5\circ}, \quad (\text{III.10})$$

which is diagonalized by the spin-weighted spheroidal harmonics. Its solution is

$$\boxed{f_{\ell mn}^\circ = \frac{2i\omega_{mn}^3}{\mathcal{D}_{\ell m\omega}}\psi_{0,\ell mn}^{5\circ} = \frac{32i\omega_{mn}^3}{\mathfrak{C}_{\ell m}\mathfrak{C}_{\ell m}}\psi_{0,\ell mn}^{5\circ}} \quad (\text{III.11})$$

where $\mathfrak{C}_{\ell m}$ is the complex Teukolsky–Starobinsky constant of Appendix B. [PZ: Check which of the two right-hand sides of (III.11) the notebook's Starobinsky $\mathcal{C}$ actually returns: the CTeukolskyStarobinsky routine returns the *complex* $\mathfrak{C} = D + 12iP\tilde{\omega}$, but fHS divides $32i\omega^3\psi_0^{5\circ}$ by it directly, which only reproduces the Schwarzschild limit $2i\omega^3/({}_2B_\ell^2 + 9M^2\omega^2)$ if the symbol actually holds $\mathfrak{C}\bar{\mathfrak{C}}$. One of the two is off; please confirm in the notebook.]

C. Held integration constants $\Phi^{n\circ}$

The four Held integration constants appearing in (I.10) are determined from $\bar{f}^\circ$ by the triangular system

$$\Phi^{3\circ} = -2\tilde{\mathbf{P}}'\bar{f}^\circ, \quad (\text{III.12a})$$

$$\Phi^{2\circ} = -2\tilde{\delta}'\tilde{\delta}\bar{f}^\circ, \quad (\text{III.12b})$$

$$\tilde{\mathbf{P}}'\Phi^{1\circ} = -\tilde{\delta}'^2\tilde{\delta}^2\bar{f}^\circ - (4\bar{\tau}^\circ\tilde{\delta} + 3\Psi^\circ)\tilde{\mathbf{P}}'\bar{f}^\circ, \quad (\text{III.12c})$$

$$\tilde{\mathbf{P}}'^2\Phi^{0\circ} = -\frac{1}{3}\tilde{\delta}'^3\tilde{\delta}^3\bar{f}^\circ - [2\bar{\tau}^\circ\tilde{\delta}'\tilde{\delta}^2 + \Psi^\circ\tilde{\delta}'\tilde{\delta} + 2\Psi^\circ\rho'^\circ]\tilde{\mathbf{P}}'\bar{f}^\circ. \quad (\text{III.12d})$$

On a Fourier mode $\tilde{\mathbf{P}}' \rightarrow -i\omega$ and the two inversions in (III.12c)–(III.12d) are divisions by $-i\omega$ and $-\omega^2$ respectively; no radial or temporal integration is required. The system is therefore purely angular, and reduces in modes to the banded linear algebra of Sec. IV. [PZ: $\Phi^{0\circ}$ and $\Phi^{3\circ}$ are called d° and e° in Vahid's notes (Eqs. 29 and 32); I have used $\Phi^{n\circ}$ throughout.]

D. The residual gauge vector ζ^a

The residual gauge vector is likewise generated by $\bar{f}^\circ$:

$$\bar{\zeta}_m^\circ = \tilde{\delta}\bar{f}^\circ, \quad (\text{III.13a})$$

$$\tilde{\mathbf{P}}' \zeta_l^\circ = -\text{Re} \{ \tilde{\delta}\tilde{\delta}\bar{f}^\circ \}, \quad (\text{III.13b})$$

$$\tilde{\mathbf{P}}' \zeta_n^\circ = -\text{Re} \left\{ \tilde{\delta}'\tilde{\delta}^3\bar{f}^\circ - \rho'^\circ \tilde{\delta}\tilde{\delta}\bar{f}^\circ - 2\tau^\circ \tilde{\delta}\tilde{\mathbf{P}}'\bar{f}^\circ - \frac{1}{2}\Omega^\circ \tilde{\delta}\tilde{\delta}\tilde{\mathbf{P}}'\bar{f}^\circ \right\}. \quad (\text{III.13c})$$

Here ζ_l° and ζ_n° are real scalars of spin weight zero, so that Re is implemented in modes through (I.16). The conjugate of (III.13a) is best written directly as $\zeta_m^\circ = \tilde{\delta}'f^\circ$ rather than obtained by conjugating $\bar{\zeta}_m^\circ$; see Sec. IV E.

E. The adjusted metric and cancellation of the growing terms

Combining the completed Hertz potential

$$\Phi^{\text{adj}} = \Phi + \Phi^{0\circ} + \frac{\Phi^{1\circ}}{\rho} + \frac{\Phi^{2\circ}}{\rho^2} + \frac{\Phi^{3\circ}}{\rho^3} \quad (\text{III.14})$$

with the residual gauge transformation gives the adjusted metric perturbation

$$\boxed{h_{ab}^{\text{adj}} = 2\text{Re} \left(\mathcal{S}^\dagger \Phi^{\text{adj}} \right)_{ab} - \mathcal{L}_\zeta g_{ab}.} \quad (\text{III.15})$$

The Lie-derivative term has the same ingoing-radiation-gauge structure as the reconstructed piece,

$$\mathcal{L}_\zeta g_{ab} l^a = 0, \quad \mathcal{L}_\zeta g_{ab} m^a \bar{m}^b = 0, \quad (\text{III.16})$$

so that the adjustment preserves the IRG conditions; its nonvanishing components are collected in Appendix D.

That (III.15) is asymptotically flat is a theorem of Ref. [13]. What the mode decomposition adds is an explicit, checkable form of it: with f° given by (III.11), $\Phi^{n\circ}$ by (III.12) and ζ^a by (III.13), the coefficients of r and of r^0 in each of h_{nn}^{adj} , h_{nm}^{adj} , $h_{n\bar{m}}^{\text{adj}}$, h_{mm}^{adj} and $h_{\bar{m}\bar{m}}^{\text{adj}}$ cancel mode by mode between the two terms of (III.15), leaving the Bondi falloff (I.8); the non-radiative components (I.9) vanish identically in the ingoing radiation gauge, as noted in Sec. II C. We verify this cancellation symbolically in Sec. V C.

F. Summary of the algorithm

- Step 1.** Solve the $s = +2$ Teukolsky equation with the point-particle source and extract the asymptotic amplitudes ${}_{+2}Z_{\ell mnk}^\infty$ at $\mathcal{I}^+$; set $\psi_{0,\ell mnk}^{5^\circ} = -{}_{+2}Z_{\ell mnk}^\infty$ (signature flip, Sec. ID).
- Step 2.** Form the seed $f_{\ell mnk}^\circ$ from (III.11): one complex division per mode.
- Step 3.** Project f° from the spheroidal ${}_{+2}S_{\ell,m}$ basis onto the spherical ${}_{+2}Y_{L,m}$ basis using the mixing coefficients (IV.2). (In Schwarzschild this step is the identity.)
- Step 4.** Build $\Phi^{0^\circ}, \dots, \Phi^{3^\circ}$ from (III.12) by repeated application of the tridiagonal operators of Sec. IV C.
- Step 5.** Assemble Φ^{adj} from (III.14); the inverse powers of ρ are the tridiagonal $\cos \theta$ matrix (IV.8), so this step widens the ℓ -support by three and requires three extra ℓ 's of padding.
- Step 6.** Build $\zeta_l^\circ, \zeta_n^\circ, \zeta_m^\circ$ from (III.13) and evaluate $\mathcal{L}_\zeta g_{ab}$ from Appendix D.
- Step 7.** Assemble h_{ab}^{adj} from (III.15) and verify the falloff.

[PZ: A figure here – a simple flow chart from ${}_{+2}Z^\infty$ to h^{adj} – would carry a lot of weight for a methods paper. Cheap to make and the referee will like it.]

IV. MODE DECOMPOSITION

A. Frequency-domain conventions

For a generic bound geodesic with radial, polar and azimuthal frequencies $(\Omega_r, \Omega_\theta, \Omega_\varphi)$ we use

$$\omega_{mnk} = m \Omega_\varphi + n \Omega_r + k \Omega_\theta, \quad (\text{IV.1})$$

reducing for circular equatorial orbits to $\omega = m \Omega_\varphi$ with $\Omega_\varphi = \sqrt{M/r_0^3}$ in Schwarzschild.

[PZ: Note that $f^\circ \propto \omega^3$, so the $\omega = 0$ modes drop out of the radiative completion entirely; the notebook returns an empty association for them. The stationary ($\omega = 0$) completion – mass and angular momentum – is handled by $\dot{g}_{ab}$ in (I.1), i.e. by the completion of Refs. [10, 11]; say so here.]

B. Spheroidal-to-spherical projection

The Teukolsky solution is naturally expressed in spin-weighted *spheroidal* harmonics, whereas the Held operators of Sec. IV C act simply on spin-weighted *spherical* harmonics. We therefore expand

$${}_sS_{\ell,m}(\theta, \varphi; a\omega) = \sum_L b_L^{(s,\ell,m)}(a\omega) {}_sY_{L,m}(\theta, \varphi), \quad (\text{IV.2})$$

so that a field with spheroidal amplitudes $X_{\ell m}$ has spherical amplitudes

$$X_{Lm}^{(Y)} = \sum_{\ell} X_{\ell m} b_L^{(s,\ell,m)}(a\omega). \quad (\text{IV.3})$$

[PZ: State how many terms you keep (NumTerms $\rightarrow$ 30 in the notebook) and the chop tolerance (10^{-25}), and give a one-line convergence statement – the coefficients fall off geometrically in $|L - \ell|$ for $a\omega \lesssim 1$. A small table or plot of $|b_L|$ vs $L - \ell$ for the worst-case $a\omega$ in your runs would settle the truncation question for a referee.]

C. Held angular operators in the ${}_sY_{\ell,m}$ basis

Write $f = \sum_{\ell m} f_{\ell m} {}_sY_{\ell,m}(\theta, \varphi)$ for a Held scalar of spin weight s . The Held angular operators are then tridiagonal in ℓ at fixed m :

$$\begin{aligned} (\tilde{\delta}f)_{\ell m} &= \left(1 - \frac{a\omega m}{\ell(\ell+1)}\right) \sqrt{\frac{(\ell-s)(\ell+s+1)}{2}} f_{\ell m} \\ &\quad + \frac{a\omega}{\ell+1} \sqrt{\frac{(\ell-m+1)(\ell+m+1)(\ell-s)(\ell-s+1)}{2(2\ell+1)(2\ell+3)}} f_{\ell+1,m} \\ &\quad - \frac{a\omega}{\ell} \sqrt{\frac{(\ell-m)(\ell+m)(\ell+s)(\ell+s+1)}{2(2\ell-1)(2\ell+1)}} f_{\ell-1,m}, \end{aligned} \quad (\text{IV.4a})$$

$$\begin{aligned} (\tilde{\delta}'f)_{\ell m} &= -\left(1 - \frac{a\omega m}{\ell(\ell+1)}\right) \sqrt{\frac{(\ell+s)(\ell-s+1)}{2}} f_{\ell m} \\ &\quad + \frac{a\omega}{\ell+1} \sqrt{\frac{(\ell-m+1)(\ell+m+1)(\ell+s)(\ell+s+1)}{2(2\ell+1)(2\ell+3)}} f_{\ell+1,m} \\ &\quad - \frac{a\omega}{\ell} \sqrt{\frac{(\ell-m)(\ell+m)(\ell-s)(\ell-s+1)}{2(2\ell-1)(2\ell+1)}} f_{\ell-1,m}, \end{aligned} \quad (\text{IV.4b})$$

while the multiplication operators by the background Held scalars are

$$\begin{aligned}
(\tau^\circ f)_{\ell m} = & -\frac{ia m}{\ell(\ell+1)} \sqrt{\frac{(\ell-s)(\ell+s+1)}{2}} f_{\ell m} \\
& + \frac{ia}{\ell+1} \sqrt{\frac{(\ell-m+1)(\ell+m+1)(\ell-s)(\ell-s+1)}{2(2\ell+1)(2\ell+3)}} f_{\ell+1,m} \\
& - \frac{ia}{\ell} \sqrt{\frac{(\ell-m)(\ell+m)(\ell+s)(\ell+s+1)}{2(2\ell-1)(2\ell+1)}} f_{\ell-1,m},
\end{aligned} \tag{IV.5a}$$

$$\begin{aligned}
(\bar{\tau}^\circ f)_{\ell m} = & +\frac{ia m}{\ell(\ell+1)} \sqrt{\frac{(\ell+s)(\ell-s+1)}{2}} f_{\ell m} \\
& + \frac{ia}{\ell+1} \sqrt{\frac{(\ell-m+1)(\ell+m+1)(\ell+s)(\ell+s+1)}{2(2\ell+1)(2\ell+3)}} f_{\ell+1,m} \\
& - \frac{ia}{\ell} \sqrt{\frac{(\ell-m)(\ell+m)(\ell-s)(\ell-s+1)}{2(2\ell-1)(2\ell+1)}} f_{\ell-1,m},
\end{aligned} \tag{IV.5b}$$

$$\begin{aligned}
(\Omega^\circ f)_{\ell m} = & +\frac{2ia m s}{\ell(\ell+1)} f_{\ell m} - \frac{2ia}{\ell+1} \sqrt{\frac{(\ell-m+1)(\ell+m+1)(\ell-s+1)(\ell+s+1)}{(2\ell+1)(2\ell+3)}} f_{\ell+1,m} \\
& - \frac{2ia}{\ell} \sqrt{\frac{(\ell-m)(\ell+m)(\ell-s)(\ell+s)}{(2\ell-1)(2\ell+1)}} f_{\ell-1,m},
\end{aligned} \tag{IV.5c}$$

$$(\tilde{\mathbf{P}}' f)_{\ell m} = -i\omega f_{\ell m}. \tag{IV.5d}$$

The $\ell = 0$ entries degenerate and are given separately in Appendix A. Equation (IV.5c) is simply $\Omega^\circ = -2ia \mathbf{C}$ written out, with $\mathbf{C}$ the $\cos \theta$ matrix (IV.6); it is quoted here only so that all five operators appear together.

At $a = 0$ every operator in (IV.4) collapses to its diagonal and the multiplication operators (IV.5) vanish identically, so the Schwarzschild construction is completely decoupled in ℓ — which is why the closed-form results of Sec. V exist at all.

[PZ: **Discrepancy in Ω° — please check.** Equation (IV.5c) above is what $\Omega^\circ = -2ia \cos \theta$ requires. The matrix elements coded as $\Omega\mathbf{H}[\mathbf{a}][\mathbf{f}][1,\mathbf{m}]$ in the notebooks do *not* agree with it. Comparing entry by entry (any s, ℓ, m gives the same ratios):

entry	(notebook)/(Eq. (IV.5c))
diagonal, $f_{\ell m}$	-1
upper, $f_{\ell+1,m}$	$-1/\sqrt{2}$
lower, $f_{\ell-1,m}$	$+1/\sqrt{2}$

No overall normalisation reconciles these, so this is not a convention difference. The cleanest diagnostic: Ω° has spin weight zero, so it is multiplication by a function and its matrix in an orthonormal basis must be *symmetric*. The notebook's is *antisymmetric* — the $f_{\ell+1,m}$ and $f_{\ell-1,m}$

coefficients have equal magnitude and opposite sign — which no multiplication operator can be. It looks as though Ω° was implemented by copying the three-term template used for $\tilde{\delta}$ and τ° , which are genuinely derivative- and spin-raising-type and do carry that relative minus. Impact: Ω° enters only (III.13c) and the $\tau^\circ, \bar{\tau}^\circ, \Omega^\circ$ terms of Appendix D, all of which vanish at $a = 0$ — so the Schwarzschild results of Sec. V are unaffected, but the Kerr gauge vector and $\mathcal{L}_\zeta g_{ab}$ are.]

[PZ: **And a related question about τ° .** The notebook's τ° carries the same opposite-sign pattern between its $f_{\ell\pm 1, m}$ entries. For τ° that is not immediately damning, since it maps the ${}_s Y_\ell$ basis to the ${}_{s+1} Y_\ell$ basis and the relevant condition is a transpose relation against $\bar{\tau}^\circ$ rather than plain symmetry. But given the Ω° problem it is worth checking explicitly, e.g. by verifying $\langle {}_{s+1} Y_{\ell, m}, \tau^\circ {}_s Y_{\ell', m} \rangle = \overline{\langle {}_s Y_{\ell', m}, \bar{\tau}^\circ {}_{s+1} Y_{\ell, m} \rangle}$ once you have the closed form of τ° for (I.12).]

D. Inverse powers of ρ and the $\cos \theta$ coupling matrix

In Kerr, $\rho^{-1} = -(r - ia \cos \theta)$ is not a multiple of the identity in ℓ -space. Using

$$\cos \theta {}_s Y_{\ell, m} = c_{\ell+1} {}_s Y_{\ell+1, m} + b_\ell {}_s Y_{\ell, m} + c_\ell {}_s Y_{\ell-1, m}, \quad (\text{IV.6})$$

with

$$c_\ell(s, m) = \sqrt{\frac{(\ell^2 - m^2)(\ell^2 - s^2)}{\ell^2(4\ell^2 - 1)}}, \quad b_\ell(s, m) = -\frac{s m}{\ell(\ell + 1)}, \quad (\text{IV.7})$$

the operator ρ^{-1} is represented by the tridiagonal matrix

$$\mathbf{R} = -r \mathbb{K} + ia \mathbf{C}, \quad \mathbf{C}_{L\ell} = \text{matrix of (IV.6)}, \quad (\text{IV.8})$$

and the adjusted Hertz potential (III.14) is, as a vector in ℓ at fixed (m, n, k) ,

$$\Phi^{\text{adj}} = \Phi^{0^\circ} + \mathbf{R} \Phi^{1^\circ} + \mathbf{R}^2 \Phi^{2^\circ} + \mathbf{R}^3 \Phi^{3^\circ}. \quad (\text{IV.9})$$

Because $\mathbf{R}$ is tridiagonal, an output at multipole L requires input up to $L + 3$; the implementation therefore pads the ℓ -range by three. [PZ: The notebook pads by 3 for Φ^{adj} and by "LPad" = 8 for ζ . Say why the ζ padding is larger (more nested operators), and quote the observed truncation error for the padding you actually use.]

E. Reality and conjugation in mode space

Conjugation is the single most error-prone step in an implementation of Sec. III, and we restate the rule here in the form in which it is applied. For a field of spin weight s with ${}_s Y_{\ell, m}$

coefficients $X_{\ell mnk}$, the conjugate field has spin weight $-s$ and $_{-s}Y_{\ell,m}$ coefficients

$$(\bar{X})_{\ell mnk} = (-1)^{s+m} \overline{X_{\ell,-m,-n,-k}}, \quad (\text{IV.10})$$

the reflection $(m, n, k) \rightarrow (-m, -n, -k)$ supplying the frequency flip $\omega_{mnk} \rightarrow -\omega_{mnk}$. Thus $\bar{\Phi}$ at mode (ℓmnk) is built from Φ at $(\ell, -m, -n, -k)$, and similarly for $\bar{f}^\circ$.

Two cautions follow. First, the sign $(-1)^s$ is unity for all of $\psi_0^{5^\circ}$, f° and Φ^{n° , so it can be omitted throughout the Hertz-potential sector without consequence; it is *not* unity for ζ_m° , for the intermediate chains $\tilde{\delta}\bar{f}^\circ$ and $\tilde{\delta}'\tilde{\delta}^2\bar{f}^\circ$, or for the nm and $n\bar{m}$ metric components (Table I). Second, where a conjugate quantity satisfies a defining equation of its own it is cheaper and safer to use it: we obtain ζ_m° from

$$\zeta_m^\circ = \tilde{\delta}' f^\circ, \quad (\text{IV.11})$$

the GHP conjugate of (III.13a), rather than by applying (IV.10) to $\bar{\zeta}_m^\circ$.

[PZ: **Please check two things in the notebooks.** (a) `BondiGauge_Adjusted.nb` defines `cconj[expr] := ComplexExpand[Conjugate[expr]] /. {ω → -ω}`, i.e. the frequency flip is inside `cconj`; the generic notebook defines `cconj` *without* the flip (it has a separate `cconj$ω` that includes it) and carries the flip instead through the $(-m, -n, -k)$ argument reflection. Both are self-consistent, but `cconjSch` differs between the two notebooks in exactly this way, and any expression copied from one into the other will be wrong by $\omega \rightarrow -\omega$. (b) The generic notebook defines both `cconjMode` (with $(-1)^m$) and `cconjMode$s` (with $(-1)^{m+s}$). Only the latter is correct in general; confirm that every odd-spin-weight use calls the `$s` variant, or that the missing $(-1)^s$ has been inserted by hand at the call site, as it appears to have been in `ReSdagModePhiOps["nm"]` and `LieZetagComps["nmb"]`.]

V. SCHWARZSCHILD IMPLEMENTATION

A. Closed-form mode formulas

In Schwarzschild the spheroidal harmonics reduce to spherical harmonics and the mixing coefficients in (IV.2) become $\delta_{L\ell}$. The angular constants reduce to

$${}_2B_\ell = \frac{1}{4} \frac{(\ell+2)!}{(\ell-2)!} = \frac{1}{4}(\ell+2)(\ell+1)\ell(\ell-1) = \frac{\Lambda_\ell}{4}, \quad {}_1B_\ell = \frac{1}{2} \frac{(\ell+1)!}{(\ell-1)!} = \frac{\ell(\ell+1)}{2}, \quad (\text{V.1})$$

with $\Lambda_\ell := (\ell - 1)\ell(\ell + 1)(\ell + 2)$, and the denominator (III.7) becomes

$$\mathcal{D}_{\ell m \omega} \longrightarrow {}_2B_\ell^2 + 9M^2\omega^2 = \frac{\Lambda_\ell^2}{16} + 9M^2\omega^2, \quad (\text{V.2})$$

which we continue to denote $\mathcal{D}_{\ell m \omega}$. The seed and the four Held integration constants are then

$$f_{\ell m}^\circ(\omega) = \frac{2i\omega^3}{\mathcal{D}_{\ell m \omega}} \psi_{0, \ell m}^{5^\circ}(\omega), \quad (\text{V.3a})$$

$$\Phi_{\ell m}^{0^\circ}(\omega) = \frac{-\frac{2}{3}i\omega(-1)^m}{\mathcal{D}_{\ell m \omega}} \left[{}_2B_{\ell 1}B_\ell - 3iM\omega(1 + \sqrt{{}_2B_\ell}) \right] \bar{\psi}_{0, \ell, -m}^{5^\circ}(-\omega), \quad (\text{V.3b})$$

$$\Phi_{\ell m}^{1^\circ}(\omega) = \frac{2\omega^2(-1)^m}{\mathcal{D}_{\ell m \omega}} [{}_2B_\ell - 3iM\omega] \bar{\psi}_{0, \ell, -m}^{5^\circ}(-\omega), \quad (\text{V.3c})$$

$$\Phi_{\ell m}^{2^\circ}(\omega) = \frac{4i\omega^3 {}_2B_{\ell/1}B_\ell(-1)^m}{\mathcal{D}_{\ell m \omega}} \bar{\psi}_{0, \ell, -m}^{5^\circ}(-\omega), \quad (\text{V.3d})$$

$$\Phi_{\ell m}^{3^\circ}(\omega) = \frac{-4\omega^4(-1)^m}{\mathcal{D}_{\ell m \omega}} \bar{\psi}_{0, \ell, -m}^{5^\circ}(-\omega), \quad (\text{V.3e})$$

and the residual gauge vector is

$$\zeta_{m, \ell m}^\circ(\omega) = \frac{-2i\omega^3 \sqrt{{}_2B_{\ell/1}B_\ell}}{\mathcal{D}_{\ell m \omega}} \psi_{0, \ell m}^{5^\circ}(\omega), \quad (\text{V.4a})$$

$$\zeta_{l, \ell m}^\circ(\omega) = \frac{\omega^2 \sqrt{{}_2B_\ell}}{\mathcal{D}_{\ell m \omega}} \psi_{0, \ell m}^{5^\circ}(\omega) + \text{c.c.}, \quad (\text{V.4b})$$

$$\zeta_{n, \ell m}^\circ(\omega) = \frac{-\omega^2 \sqrt{{}_2B_\ell}}{\mathcal{D}_{\ell m \omega}} \left({}_1B_\ell + \frac{1}{2} \right) \psi_{0, \ell m}^{5^\circ}(\omega) + \text{c.c.}, \quad (\text{V.4c})$$

where “+ c.c.” denotes the mode-space real part (I.16), i.e. the addition of $(-1)^m \bar{\psi}_{0, \ell, -m}^{5^\circ}(-\omega)$ inside the same prefactor. **[PZ: Discrepancy to resolve. Equation (V.3b) uses ${}_2B_{\ell 1}B_\ell = \frac{1}{8}(\ell + 2)(\ell + 1)^2\ell^2(\ell - 1)$, following Vahid's notes Eq. (29). BondiGauge_Adjusted.nb instead codes $\frac{1}{8}(\ell + 2)(\ell + 1)\ell^2(\ell - 1)^2$, i.e. $(\ell + 1)^2(\ell - 1)$ replaced by $(\ell + 1)(\ell - 1)^2$ – these differ for every $\ell \geq 2$. Everything else in (V.3)–(V.4) agrees with the notebook up to the uniform factor of 2 in the Φ^{n° noted in Sec. I D. The cleanest test is whether the $O(r)$ cancellation in Sec. V C still holds with the notes' coefficient.]**

Note that $\sqrt{{}_2B_\ell} = \frac{1}{2}\sqrt{\Lambda_\ell}$ and ${}_2B_{\ell/1}B_\ell = \frac{1}{2}(\ell + 2)(\ell - 1)$, which is the form in which these factors appear in the implementation.

B. Reconstruction and Lie-derivative operators in modes

The explicit Schwarzschild mode expressions for $(\mathcal{S}^\dagger \Phi)_{ab}$ and $(\mathcal{L}_\zeta g)_{ab}$ are collected in Appendices C and D. Both are diagonal in ℓ and depend on it only through Λ_ℓ .

C. Verification of the falloff

[PZ: Table here. Columns: NP component; coefficient of r ; coefficient of r^0 ; leading surviving power. Rows: nn , nm , $n\bar{m}$, mm , $\bar{m}\bar{m}$, for say $(\ell, m) = (2, 2), (2, 0), (3, 2), (3, -3)$. Your notebook already computes exactly these via `Coefficient[hAdjustment[comp][1,m][r],r].`]

An independent check is provided by the “old” and “new” implementations of $\mathcal{S}^\dagger$ in the accompanying notebooks: one applies the operator directly in closed form, the other builds it from the Held operators of Sec. IV C and reduces at the end. The two agree to machine precision for all modes tested. [PZ: Quote the actual agreement level (the notebook does `SetPrecision[Chop[checkTable],32]`). This is a good, cheap validation to report.]

D. Results for circular equatorial orbits

[PZ: TODO – numerics. Suggested content: (i) a plot of the NP components of h^{ret} and h^{adj} versus r out to $r \sim 10^3 M$, showing the $O(r)$ growth removed; (ii) a convergence plot in ℓ_{max} ; (iii) the orbital parameters used ($r_0 = 10M$ in the notebook) and the source of the ψ_0^{50} data (`psi0LargeRdata_lmax40_rOutbd1212.wxf`). Please also state how ψ_0^{50} was extracted – a large- r fit to the radial solution, or the $\mathcal{J}$ amplitude directly? The notebook filename suggests a fit, which is worth documenting since it sets the achievable precision.]

VI. KERR IMPLEMENTATION

A. What changes relative to Schwarzschild

In Schwarzschild every step of Sec. III F is diagonal in ℓ . In Kerr, exactly three ingredients break that diagonality, and they are the whole of the additional work:

- (a) the input Teukolsky solution is expanded in spin-weighted *spheroidal* harmonics, whose overlap with the spherical basis spreads each ℓ over a band of L (Sec. VI C);

- (b) the Held angular operators $\tilde{\delta}, \tilde{\delta}'$ acquire off-diagonal $a\omega$ terms, and the multiplication operators $\tau^\circ, \bar{\tau}^\circ, \Omega^\circ$ — absent in Schwarzschild — are themselves tridiagonal (Sec. VID);
- (c) $\rho^{-1} = -(r - ia \cos \theta)$ is no longer proportional to the identity, so assembling Φ^{adj} from the Φ^{n° mixes three further neighbours (Sec. VIE).

Everything else — the algebraic inversion for f° , the triangular system for the Φ^{n° , the $\tilde{\mathbf{P}}' \rightarrow -i\omega$ replacement — is unchanged. In particular every operator remains *diagonal in m* , so the completion commutes with an m -mode decomposition of the field; the m -mode scheme of Ref. [17] may be used without modification.

B. Asymptotic amplitudes for generic bound orbits

For a generic bound geodesic the frequencies are (IV.1) and the amplitudes ${}_{+2}Z_{\ell mnk}^\infty$ are computed mode by mode over the lattice

$$|s| \leq \ell \leq \ell_{\max}, \quad |m| \leq \ell, \quad |n| \leq n_{\max}, \quad |k| \leq k_{\max}. \quad (\text{VI.1})$$

Only the canonical half of the lattice,

$$m > 0 \quad \text{or} \quad (m = 0 \text{ and } [n > 0 \text{ or } (n = 0 \text{ and } k \geq 0)]), \quad (\text{VI.2})$$

need be computed directly; the remaining amplitudes follow from the reflection symmetry (III.8). This halves the dominant cost of the calculation, which is the set of Teukolsky radial solves and not the completion itself (Sec. VIG). **[PZ: State the (n, k) truncation used in production and how it was chosen (residual amplitude threshold?). The notebook's exploratory values are $\ell_{\max} = 4$, $n_{\max} = k_{\max} = 4$, $p = 10M$, $e = 0.1$, $\cos \chi = \cos(\pi/4)$, $a = 0.667M$. Worth also describing the per- ℓ caching: the implementation writes one file per ℓ as it goes and merges into a global amplitude table, so a run is restartable — a practical point that belongs in the text if you want others to reproduce this.]**

C. Spheroidal-to-spherical mixing

The seed f° of (III.11) is obtained in the spheroidal basis, one amplitude per (ℓmnk) , exactly as in Schwarzschild. It is then projected onto the spherical basis using (IV.2): writing

$f_{\ell mnk}^\circ$ for the spheroidal amplitude, the spherical coefficients are

$$f_{Lmnk}^\circ = \sum_{\ell} f_{\ell mnk}^\circ b_L^{(+2,\ell,m)}(a\omega_{mnk}), \quad (\text{VI.3})$$

and likewise for $\bar{f}^\circ$ with $s = -2$. Because the mixing coefficients depend on $(s, \ell, m, a\omega)$ only, they are computed once per mode and reused by every Held chain downstream; in practice this memoization dominates the efficiency of the whole completion, since the chains of Sec. VID would otherwise recompute the same expansion $O(10)$ times per mode.

Denote by N_{mix} the half-width at which the expansion is truncated,

$$b_L^{(s,\ell,m)} \text{ retained for } |L - \ell| \leq N_{\text{mix}}. \quad (\text{VI.4})$$

[PZ: Quote N_{mix} and justify it. The notebook uses "NumTerms" $\rightarrow 30$ with a chop tolerance of 10^{-25} , which is far more than needed for $a\omega \lesssim 1$; a plot of $|b_L|$ against $|L - \ell|$ at the largest $a\omega_{mnk}$ occurring in your run would let you quote a much smaller effective N_{mix} and would answer the truncation question cleanly. This also feeds directly into the padding budget of Table II.]

D. Held chains as banded operators

Each of $\tilde{\delta}$, $\tilde{\delta}'$, τ° , $\bar{\tau}^\circ$ and Ω° is tridiagonal in ℓ at fixed m by (IV.4)–(IV.5), and $\tilde{\mathbf{P}}'$ is diagonal by (IV.5d). A chain of p such operators is therefore banded with half-bandwidth p : its value at output multipole L depends on the input at $L - p, \dots, L + p$. Applying this to (III.12) and (III.13) gives the padding budget of Table II.

Two implementation points follow. First, because every operator couples $L \rightarrow \{L - 1, L, L + 1\}$, a naive recursive evaluation of a chain of length p costs 3^p leaf evaluations — 729 for Φ^{00} — whereas evaluating and caching each intermediate chain once, level by level, reduces the cost to $O(p L_{\text{max}})$. We therefore build and store the intermediate quantities

$$\tilde{\delta}\bar{f}^\circ, \quad \tilde{\delta}^2\bar{f}^\circ, \quad \tilde{\delta}^3\bar{f}^\circ, \quad \tilde{\delta}'\tilde{\delta}\bar{f}^\circ, \quad \tilde{\delta}'\tilde{\delta}^2\bar{f}^\circ, \quad \tilde{\delta}'^2\tilde{\delta}^2\bar{f}^\circ, \quad \tilde{\delta}'^3\tilde{\delta}^3\bar{f}^\circ, \quad (\text{VI.5})$$

together with $\bar{\tau}^\circ\tilde{\delta}\bar{f}^\circ$ and $\bar{\tau}^\circ\tilde{\delta}'\tilde{\delta}^2\bar{f}^\circ$, and assemble the Φ^{n0} and ζ° components from them. Second, the spin weight changes along each chain ($-2 \rightarrow -1 \rightarrow 0 \rightarrow +1$ under successive $\tilde{\delta}$), and the matrix elements (IV.4) depend on it, so the spin weight must be tracked with the intermediate rather than inferred at the end.

quantity	longest operator chain	half-bandwidth
$\Phi^{3\circ}$	$\tilde{\mathbf{p}}'$	0
$\Phi^{2\circ}$	$\tilde{\delta}'\tilde{\delta}$	2
$\Phi^{1\circ}$	$\tilde{\delta}'^2\tilde{\delta}^2$	4
$\Phi^{0\circ}$	$\tilde{\delta}'^3\tilde{\delta}^3$	6
$\bar{\zeta}_m^\circ$	$\tilde{\delta}$	1
ζ_l°	$\tilde{\delta}^2$	2
ζ_n°	$\tilde{\delta}'\tilde{\delta}^3$	4
ρ^{-n}	$\mathbf{C}^n$	$n \leq 3$
Φ^{adj}	$\max\{6, 1+4, 2+2, 3+0\}$	6

TABLE II. Half-bandwidth in ℓ of each stage of the completion, at fixed (m, n, k) . To obtain Φ^{adj} correctly for $L \leq L_{\text{out}}$ the seed $\bar{f}^\circ$ must be supplied for $\ell \leq L_{\text{out}} + 6$, and that seed must itself be assembled from spheroidal amplitudes out to $\ell \leq L_{\text{out}} + 6 + N_{\text{mix}}$.

E. Assembling Φ^{adj}

With the $\Phi^{n\circ}$ in hand as vectors over ℓ at fixed (m, n, k) , the adjusted Hertz potential follows from (IV.9). It is evaluated in Horner form,

$$\Phi^{\text{adj}} = \Phi^{0\circ} + \mathbf{R}\left(\Phi^{1\circ} + \mathbf{R}(\Phi^{2\circ} + \mathbf{R}\Phi^{3\circ})\right), \quad (\text{VI.6})$$

which requires three tridiagonal multiplications rather than six and consumes three multipoles of padding. Note that $\mathbf{R}$ carries the explicit radial dependence through $-r\mathbb{K}$, so (VI.6) is a polynomial in r of degree three with ℓ -banded matrix coefficients; Φ^{adj} and its first two radial derivatives, which are what Appendix C requires, are obtained from it analytically.

F. The gauge vector in the spherical basis

The components of ζ^a are built from the same cached chains. Two details differ from the Hertz sector. First, ζ_l° and ζ_n° are real spin-zero scalars and are assembled through the projection (I.16), which pairs the (m, n, k) and $(-m, -n, -k)$ chains; ζ_m° is instead obtained directly from (IV.11) to avoid an explicit spin-weighted conjugation. Second, the minimum

output multipole depends on the component: nearest-neighbour mixing can feed the spin-zero components down to $L = |m|$, whereas the spin- (± 1) components start at

$$L_{\min} = \begin{cases} |m|, & \text{components } \zeta_l^\circ, \zeta_n^\circ \quad (s = 0), \\ \max(1, |m|), & \text{components } \zeta_m^\circ, \bar{\zeta}_m^\circ \quad (s = \pm 1). \end{cases} \quad (\text{VI.7})$$

Finally, since $f^\circ \propto \omega^3$, modes with $\omega_{mnk} = 0$ contribute nothing to the radiative completion and are skipped. [PZ: Quote the L -padding actually used for ζ and the chop tolerance below which coefficients are discarded. Table II suggests 4 suffices for ζ_n° ; the notebook uses 8, so either the extra margin is unnecessary or I have miscounted the chain — worth checking, since it doubles the cost of this stage.]

G. Cost and truncation

At fixed (m, n, k) the completion consists of one complex division per ℓ for the seed, one spheroidal projection per ℓ , and a fixed number (≈ 20) of tridiagonal applications over the padded ℓ -range. Its cost is therefore $O(L_{\max})$ per (m, n, k) , against the $O(L_{\max})$ *Teukolsky radial solves* needed to produce the input amplitudes in the first place, each of which is more expensive by orders of magnitude. The asymptotic completion is thus essentially free relative to the calculation it corrects. [PZ: Back this with one timing number — e.g. wall-clock for the amplitude run versus wall-clock for the completion at the same $\ell_{\max}$. A single sentence with real numbers is worth a paragraph of argument here.]

H. Results for circular equatorial orbits

[PZ: TODO – numerics. Same three plots as Sec. VD, plus: (i) a convergence study in N_{mix} at fixed $\ell_{\max}$, which is the one genuinely new truncation in Kerr; (ii) a spin sweep if you have it; (iii) the $a \rightarrow 0$ limit of the Kerr pipeline compared against the closed-form Schwarzschild results of Sec. V — that is the strongest single validation available and costs nothing to run.]

VII. DISCUSSION

[PZ: TODO – see the roadmap comment in the source. Point 3 is the one to be careful about: be explicit that this paper treats the $\mathcal{J}^+$ side only, and that the ingoing (H^-) counterpart is a

separate construction.]

ACKNOWLEDGMENTS

This work made use of the Black Hole Perturbation Toolkit [16], in particular the Teukolsky, KerrGeodesics and SpinWeightedSpheroidalHarmonics packages. [PZ: Add funding acknowledgements.]

Appendix A: Held angular operators: degenerate cases and conventions

For $\ell = 0$ the expressions (IV.4)–(IV.5) degenerate to

$$(\tilde{\delta}f)_{0m} = a\omega\sqrt{\frac{s(s-1)}{6}} f_{1m}, \quad (\tilde{\delta}'f)_{0m} = a\omega\sqrt{\frac{s(s+1)}{6}} f_{1m}, \quad (\text{A.1a})$$

$$(\tau^\circ f)_{0m} = ia\sqrt{\frac{s(s-1)}{6}} f_{1m}, \quad (\bar{\tau}^\circ f)_{0m} = ia\sqrt{\frac{s(s+1)}{6}} f_{1m}, \quad (\text{A.1b})$$

$$(\Omega^\circ f)_{00} = -\frac{2ia}{\sqrt{3}} f_{10}. \quad (\text{A.1c})$$

The $\tilde{\delta}$, $\tilde{\delta}'$, τ° and $\bar{\tau}^\circ$ entries are transcribed from the `If[1==0,...]` branches of the implementation. The Ω° entry is instead the $\ell = 0$ limit of (IV.5c): only $s = 0$, $m = 0$ occurs there, and $c_1(0,0) = 1/\sqrt{3}$. [PZ: The notebook's $\ell = 0$ branch for Ω° reads $ia\sqrt{-2(s-1)(s+1)/3} f_{1,0}$, which at $s = 0$ gives $+ia\sqrt{2/3}$ rather than the $-2ia/\sqrt{3}$ above — the same disagreement flagged after (IV.5c), carried into the degenerate case. Since only $s = 0$ can occur at $\ell = 0$, the s -dependence in that branch is inert in any event. These entries matter only for the spin-zero gauge-vector components, where Kerr nearest-neighbour mixing can feed $\ell = 0, 1$; cf. the $L_{\min}$ switch (VI.7).]

Appendix B: Teukolsky–Starobinsky constants

For $s = \pm 2$ the complex Teukolsky–Starobinsky constant is

$$\mathfrak{C}_{\ell m}(a\omega) = D_{\ell m}(a\omega) + 12iP\tilde{\omega}, \quad (\text{B.1})$$

with

$$D_{\ell m} = \left[\lambda_{\text{Ch}}^2 (\lambda_{\text{Ch}} - 2)^2 + 8a\tilde{\omega}(m - a\tilde{\omega})(\lambda_{\text{Ch}} - 2)(5\lambda_{\text{Ch}} - 4) + 48(a\tilde{\omega})^2 (2(\lambda_{\text{Ch}} - 2) + 3(m - a\tilde{\omega})^2) \right]^{1/2}, \quad (\text{B.2})$$

where $\lambda_{\text{Ch}} = \lambda + s^2 + s$ and λ is the spin-weighted spheroidal eigenvalue, and $P = \pm 1$ is the parity factor of (III.5); see Ref. [18] for a careful discussion of these constants and of the conventions attached to P . In Schwarzschild $\lambda_{\text{Ch}} = \ell(\ell + 1)$, $D_\ell = \Lambda_\ell$, and

$$\mathfrak{C}\bar{\mathfrak{C}} = \Lambda_\ell^2 + 144M^2\omega^2 = 16(2B_\ell^2 + 9M^2\omega^2), \quad (\text{B.3})$$

which reproduces the Schwarzschild limit of (III.11) and fixes the normalisation $\mathfrak{C}\bar{\mathfrak{C}} = 16 \mathcal{D}_{\ell m \omega}$ used there.

Appendix C: Reconstruction operator in modes (Schwarzschild)

With $\Phi = \sum_{\ell m} \Phi_{\ell m}(r) {}_{-2}Y_{\ell, m}$ in the ingoing radiation gauge, the nonvanishing components of $\mathcal{S}^\dagger \Phi$ are

$$(\mathcal{S}^\dagger \Phi)_{nn} = -\frac{1}{r^2} \sqrt{\frac{\Lambda_\ell}{16}} \left[\Phi_{\ell m}(r) + (-1)^m \bar{\Phi}_{\ell, -m}(r) \right], \quad (\text{C.1a})$$

$$(\mathcal{S}^\dagger \Phi)_{nm} = -\frac{(-1)^m}{r^2} \sqrt{\frac{(\ell-1)(\ell+2)}{8}} \left[-2\Phi_{\ell, -m}(r) + r \Phi'_{\ell, -m}(r) \right], \quad (\text{C.1b})$$

$$(\mathcal{S}^\dagger \Phi)_{n\bar{m}} = \frac{\sqrt{(\ell-1)(\ell+2)}}{2\sqrt{2}r^2} \left[-2\Phi_{\ell m}(r) + r \Phi'_{\ell m}(r) \right], \quad (\text{C.1c})$$

$$(\mathcal{S}^\dagger \Phi)_{mm} = (-1)^m \left[\frac{1}{r} \Phi'_{\ell, -m}(r) - \frac{1}{2} \Phi''_{\ell, -m}(r) \right], \quad (\text{C.1d})$$

$$(\mathcal{S}^\dagger \Phi)_{\bar{m}\bar{m}} = \frac{1}{r} \Phi'_{\ell m}(r) - \frac{1}{2} \Phi''_{\ell m}(r), \quad (\text{C.1e})$$

with $(\mathcal{S}^\dagger \Phi)_{ll} = (\mathcal{S}^\dagger \Phi)_{ln} = (\mathcal{S}^\dagger \Phi)_{lm} = (\mathcal{S}^\dagger \Phi)_{l\bar{m}} = (\mathcal{S}^\dagger \Phi)_{m\bar{m}} = 0$. The signs multiplying the conjugated amplitudes in (C.1a)–(C.1e) are not independent choices: each is the factor $(-1)^{s+m}$ of the conjugation rule (I.15), evaluated at the spin weight of the component being formed. Thus (C.1a) and (C.1d) carry $(-1)^m$, because h_{nn} has $s = 0$ and h_{mm} has $s = +2$, while (C.1b) carries $-(-1)^m$, because h_{nm} has $s = +1$. **[PZ: Two things to check here. (a) Overall factor. (C.1a) already contains the combination $\Phi + (-1)^m \bar{\Phi}$, which is twice the mode-space real part (I.16), yet the notebook multiplies the whole set by a further factor of 2. Fix the convention once and propagate it; the falloff cancellation is insensitive to an overall factor but the amplitudes are not. (b) Which component carries the minus. In `ReSdagModePhiOps` the $-(-1)^m$ sits on "nm" and "nmb" is the primitive expression, whereas in `LieZetagComps` it is the other way round — "nm" is primitive and "nmb" carries the sign. Both are internally consistent,**

but they can only be added together in (III.15) if the two conventions for which of m^a , $\bar{m}^a$ is contracted first agree. Worth verifying explicitly, since an error here would flip the sign of one $O(r)$ term and would not show up in the nn or mm channels.]

Appendix D: Lie derivative of the metric along ζ^a

With $\mathcal{L}_\zeta g_{ab} l^a = 0$ and $\mathcal{L}_\zeta g_{ab} m^a \bar{m}^b = 0$, the remaining components are

$$\mathcal{L}_\zeta g_{ab} m^a m^b = -2\bar{\rho} \tilde{\partial} \tilde{\partial} \zeta_l^\circ + 2\tilde{\partial}' \tilde{\partial} f^\circ + 2\Omega^\circ \tilde{\mathbf{P}}' f^\circ + 4\bar{\rho} \tau^\circ \tilde{\partial}' f^\circ + 8\rho'^\circ f^\circ, \quad (\text{D.1})$$

$$\begin{aligned} \mathcal{L}_\zeta g_{ab} n^a m^b &= \bar{\rho} \tilde{\partial}' \tilde{\partial} \tilde{\partial} \zeta_l^\circ - 2\rho \bar{\rho} \bar{\tau}^\circ \tilde{\partial} \tilde{\partial} \zeta_l^\circ + \rho(\rho + \bar{\rho}) \Psi^\circ \tilde{\partial} \zeta_l^\circ \\ &\quad - \bar{\rho} \Omega^\circ \tilde{\partial}' \tilde{\partial}' \tilde{\partial} f^\circ + 2\rho \bar{\tau}^\circ \tilde{\partial}' \tilde{\partial} f^\circ \\ &\quad + \frac{1}{\bar{\rho}} \left(-2\bar{\rho} \Omega^{\circ 2} \tilde{\partial}' \tilde{\mathbf{P}}' f^\circ + 2(\rho + \bar{\rho}) \Omega^\circ \bar{\tau}^\circ \tilde{\mathbf{P}}' f^\circ - 2\bar{\rho} \tau^\circ \tilde{\partial}' \tilde{\partial}' f^\circ \right. \\ &\quad \left. - (\rho^2 \Omega^\circ + \rho) \Psi^\circ \tilde{\partial}' f^\circ \right) \\ &\quad + \bar{\rho} \bar{\Psi}^\circ \tilde{\partial}' f^\circ - 6\bar{\rho} \Omega^\circ \rho'^\circ \tilde{\partial}' f^\circ + 4\rho \bar{\rho} \tau^\circ \bar{\tau}^\circ \tilde{\partial}' f^\circ + 8\rho \rho'^\circ \bar{\tau}^\circ f^\circ, \end{aligned} \quad (\text{D.2})$$

and

$$\begin{aligned} \mathcal{L}_\zeta g_{ab} n^a n^b &= \text{Re} \left[4\rho \bar{\rho} \bar{\tau}^\circ \tilde{\partial}' \tilde{\partial} \tilde{\partial} \zeta_l^\circ - 4\rho^2 \bar{\rho} \bar{\tau}^{\circ 2} \tilde{\partial} \tilde{\partial} \zeta_l^\circ + 2\rho^2 \Psi^\circ \tilde{\partial}' \tilde{\partial} \zeta_l^\circ + 4\rho^2 \bar{\rho} \Psi^\circ \bar{\tau}^\circ \tilde{\partial} \zeta_l^\circ \right. \\ &\quad \left. - 2\tilde{\partial}' \tilde{\partial}' \tilde{\partial}' \tilde{\partial} f^\circ - \frac{2}{\rho} \left(4\Omega^\circ \tilde{\partial}' \tilde{\partial}' \tilde{\mathbf{P}}' f^\circ - 4\rho \bar{\rho} \Omega^\circ \bar{\tau}^\circ \tilde{\partial}' \tilde{\partial}' \tilde{\partial} f^\circ \right. \right. \\ &\quad \left. \left. + 4(4 + \bar{\rho} \Omega^\circ - \rho \bar{\rho} \Omega^{\circ 2}) \bar{\tau}^\circ \tilde{\partial}' \tilde{\mathbf{P}}' f^\circ \right) \right. \\ &\quad \left. - (8\rho'^\circ + 3\rho \Psi^\circ + 3\bar{\rho} \bar{\Psi}^\circ + 2\rho^2 \Omega^{\circ 2} \Psi^\circ + 8\rho \bar{\rho} \tau^\circ \bar{\tau}^\circ) \tilde{\partial}' \tilde{\partial}' f^\circ \right. \\ &\quad \left. + 4\rho^2 \bar{\tau}^{\circ 2} \tilde{\partial}' \tilde{\partial} f^\circ + 4\rho(\rho + 2\bar{\rho}) \Omega^\circ \bar{\tau}^{\circ 2} \tilde{\mathbf{P}}' f^\circ + 8\rho^2 \bar{\rho} \tau^\circ \bar{\tau}^{\circ 2} \tilde{\partial}' f^\circ \right. \\ &\quad \left. + 4\rho \bar{\rho} \bar{\Psi}^\circ \bar{\tau}^{\circ 2} \tilde{\partial}' f^\circ - 24\rho \bar{\rho} \Omega^\circ \rho'^\circ \bar{\tau}^\circ \tilde{\partial}' f^\circ + 16\rho^2 \rho'^\circ \bar{\tau}^{\circ 2} f^\circ \right]. \end{aligned} \quad (\text{D.3})$$

The remaining components follow by conjugation, $\mathcal{L}_\zeta g_{ab} n^a \bar{m}^b = \overline{\mathcal{L}_\zeta g_{ab} n^a m^b}$ and $\mathcal{L}_\zeta g_{ab} \bar{m}^a \bar{m}^b = \overline{\mathcal{L}_\zeta g_{ab} m^a m^b}$, which in mode space are (I.15) at $s = +1$ and $s = +2$ respectively, i.e.

$$(\mathcal{L}_\zeta g)_{n\bar{m}, \ell m} = -(-1)^m \overline{(\mathcal{L}_\zeta g)_{nm, \ell, -m}}, \quad (\mathcal{L}_\zeta g)_{\bar{m}\bar{m}, \ell m} = (-1)^m \overline{(\mathcal{L}_\zeta g)_{mm, \ell, -m}}. \quad (\text{D.4})$$

The relative minus sign between the two is the $(-1)^s$ of (I.15) and is not optional. **[PZ: Transcription warning.** (D.1)–(D.3) are transcribed from Eqs. (18)–(20) of `metric-reconst.pdf`, where the $n^a n^b$ expression in particular is typeset across a page break and the grouping of the

$1/\rho$ bracket in (D.3) is ambiguous. Please check them against LieZetagComps in the notebook (which is the $a \rightarrow 0$ specialisation, so it will validate the Ψ° , ρ'° and ζ_l° terms but *not* the τ° , $\bar{\tau}^\circ$, Ω° terms). A Kerr-side check against Hollands–Toomani would be worth doing before submission.]

[PZ: The $a \rightarrow 0$ specialisation actually used in the Schwarzschild numerics is considerably shorter; consider giving it as a separate displayed set, since a reader reproducing Sec. V needs only that. I have not transcribed it here to avoid duplicating an expression that is already long, but it is the "nn"/"nm"/"nmb"/"mm"/"mbmb" entries of LieZetagComps with $\rho = \bar{\rho} = -1/r$, $a = 0$.]

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